

Human Structure Learning Outcomes

Embryology Of The Reproductive & Renal Systems

Week 202

Reproductive

Learning outcome

Name the parts of the developing urinary system, and identify the processes of formation of this system. ☐

Identify the parts of the developing reproductive systems in male and female embryos, and identify the processes by which the adult structures develop. ☐

Session learning outcome:

To be able to name the parts of the developing genito-urinary system and identify the processes by which the system forms and matures.

Human Structure Learning Outcomes

Oral Cavity And Salivary Glands

Week 204

Respiratory

Learning outcome

Demonstrate the major features and boundaries of the oral cavity and summarise its sensory innervation. ☐

Identify the parotid, submandibular and sublingual salivary glands, and their locations in the living, the courses of their ducts into the oral cavity, their autonomic secretomotor innervation, lymph drainage routes, and their secretions. ☐

Appreciate the narrow points of the ducts in relation to salivary stone impaction. ☐

List the structures within the parotid gland. ☐

Describe the structure of the tongue including the papillae. ☐

List the intrinsic and extrinsic muscles of the tongue, describing their attachments and innervations. ☐

Explain the sensory innervation (general and taste sensation) of the tongue. ☐

Explain the deviation of the tongue after hypoglossal nerve injuries. ☐

Describe the otic ganglion. ☐

Session learning outcome:

To be able to identify the structures within the oral cavity including the salivary glands and tongue and link them to function.

Human Structure Learning Outcomes

Large Intestine And Appendix

Week 206

Gastrointestinal

Learning outcome

Identify the caecum, vermiform appendix, ascending, transverse, descending and sigmoid colon, rectum and anal canal, and be able to name the appropriate arterial and nervous supplies and venous and lymphatic drainage. Note their positions in the living and that the colic arteries form an anastomosis. ☐

Describe the anatomy of the appendix and the significance of its variations in relation to appendicitis. ☐

Describe the anatomy of the ischio-anal/rectal fossa and explain its potential involvement in abscesses, anal glands and fissures. ☐

Describe the anatomy of the sigmoid colon and rectum and their anatomical relationships. ☐

Identify the internal anal sphincter and the insertion of levator ani into the lateral wall of the anal canal and the pubo-rectalis part of the muscle which passes round the posterior wall. ☐

Explain the functional anatomy of the anal sphincters and their role in faecal continence. ☐

Explain that most of the anal canal is in the perineum, not the pelvis. ☐

Describe the blood supply and venous drainage of the distal bowel; the supply from the superior rectal (inferior mesenteric), middle rectal (internal iliac) and inferior rectal arteries (from pudendal to anal canal only), and portal-systemic venous anastomosis. ☐

Describe the vascular anal cushions and explain their role in continence. ☐

Recognise the regions of the large intestine on radiographs. ☐

Session learning outcome:

To be able to identify the large intestine, appendix and link them to function and likely pathology.

Human Structure Learning Outcomes

Nose And Sinuses

Week 207

Respiratory

Learning outcome

Name the parts of the external nose. ☐

Name the bones that form the floor, roof and lateral walls of the nasal cavity and locate them on a model of a skull. ☐

Describe the major landmarks on the lateral wall of the nose and the major arteries that supply the lateral wall and the nasal septum (in relation to nosebleeds). ☐

Label on a diagram: superior, middle and inferior conchae, spheno-ethmoidal recess, superior, middle and inferior meatus. ☐

Locate paranasal air sinuses (frontal, maxillary, ethmoidal and sphenoidal). ☐

Describe the drainage of the air sinuses and the nasolacrimal ducts and their relationships to the nasal cavities. ☐

Explain the innervation of the paranasal sinuses in relation to referred pain. ☐

Discuss the nerve supply of the nasal cavity, including the olfactory nerves and the nerves of general sensation. ☐

Recognise structures of the nasal cavities, air cavities and bones of the face on radiographs. ☐

Session learning outcome:

To be able to identify the structures of the nose and sinuses and link them to function.

Human Structure Learning Outcomes

Hearing And Balance

Week 208

Nervous

Learning outcome

Identify the pharyngeal arches, pouches and clefts, and recall the major derivatives of each. ☐

Identify the parts of the external ear. ☐

Recognise the parts of the external, middle (tympanic cavity) and internal ear (i.e. including the external auditory meatus, tympanic membrane, semicircular canals, vestibulocochlear and facial nerves, malleus, incus, stapes, tensor tympani muscle, stapedius muscle and the internal carotid artery). ☐

Locate the chorda tympani nerve and describe its path and functions. ☐

Locate the tympanic plexus. ☐

Describe the anatomy of the auditory tube. ☐

Describe the development of the paranasal sinuses. ☐

Describe how the auditory tube changes with age and what effect this has on function. ☐

Locate the lamina papyracea and the ethmoid sinuses. ☐

Identify the stylomastoid foramen and note how the mastoid process protects the facial nerve. ☐

Identify the three main segments (intracranial, infratemporal and extratemporal) of the facial nerve. ☐

Identify the branches of the facial nerve and describe their functions. ☐

Identify sites of potential accidental and surgical trauma to the facial nerve. ☐

Identify sites where disease may affect the facial nerve. ☐

Relate form to function of the outer, middle and inner ear. ☐

Describe how the anatomy and functions of the macula and ampulla differ and what they detect. ☐

Describe the vestibulo-ocular reflex. ☐

Session learning outcome:

Diagnose and manage patients presenting with hearing loss, tinnitus, ear discharge, ear pain or vertigo.

Human Structure Learning Outcomes

Blood Supply To The Brain

Week 209

Nervous

Learning outcome

Identify and describe the paths of the vertebral and internal carotid arteries. ☐

Identify the blood supply to the brain stem and cerebrum and link to the functional outcome of interruption of the blood supply in different arterial territories. ☐

Describe the distribution of the basilar artery to the pons, membranous labyrinth and cerebellum. ☐

Identify all the components of the circle of Willis (circulus arteriosus) and make a labelled diagram of it. ☐

Describe the course of the middle meningeal artery and explain the importance of its relationship to the skull bones and interior of the cranium. ☐

Identify the main structures involved in venous drainage of the brain and link to the symptoms and signs of cerebral venous sinus thrombosis. ☐

Explain how the structure of the venous sinuses contribute to the development of subdural haemorrhage, particularly with relevance to the lucid interval. ☐

Differentiate between subdural and extradural haemorrhage based on appearance and location on radiographs. ☐

Session learning outcome:

To be able to identify the blood supply of the brain and link to functional outcome of interruption.

Human Structure Learning Outcomes

Lower Limb

Week 210

Musculoskeletal

Learning outcome

Identify and side the femur, describing key features on bones and radiographs. ☐

Describe what structures may be damaged by a fracture of the femoral neck or dislocation of the hip and explain the functional consequences of these injuries. ☐

Locate and describe the anatomical relationships of the great saphenous vein, fascia lata, iliotibial tract, femoral sheath and its contents (and adjacent femoral nerve), obturator nerve, sciatic nerve, femoral nerve, common peroneal nerve, tibial nerve, popliteal fossa and key vessels within, and identify the main muscles and muscle groups that the nerves supply. ☐

Describe the anatomical basis (nerve root or peripheral nerve) for loss of movements and reflexes at the knee and ankle resulting from spinal injuries, disc lesions and common peripheral nerve injuries. ☐

Identify the sacral plexus and its branches lying on piriformis. ☐

Locate the sciatic nerve, which is the large lateral terminal branch of the sacral plexus, and trace it to the greater sciatic foramen and the gluteal region. ☐

Describe the coccygeal plexus and the muscles it supplies. ☐

Identify the obturator nerve and explain that its course is from the medial margin of psoas muscle to the obturator foramen. ☐

Describe the courses of the principal veins of the leg. ☐

Explain the function of the connection of the superficial to the deep veins via the perforator veins and link to the development of varicose veins. ☐

Explain the mechanisms in the lower limb responsible for venous return to the heart (the muscular venous pump). ☐

Describe the boundaries of the femoral triangle and the anatomical relationships of the femoral nerve, artery, vein and lymph nodes to each other and the inguinal ligament, with particular regard to arterial blood sampling and catheter placement. ☐

Demonstrate the courses of the major arteries that supply the hip, gluteal region, thigh, leg, ankle and foot. ☐

Explain the functional significance of anastomoses between branches of the arteries at the hip and knee. ☐

Demonstrate the locations at which the femoral, popliteal, dorsalis pedis and posterior tibial pulses may be felt and predict the effects of arterial pathology. ☐

Describe the lymphatic drainage of the lower limb and its relationship to tumour spread. ☐

Interpret standard diagnostic images of the lower limb and be able to recognise common abnormalities. ☐

Human Structure Learning Outcomes

Session learning outcome:

To be able to identify the vascular structures of the lower limb and link them to function.

Human Structure Learning Outcomes

Ankle

Week 211

Cardiovascular

Learning outcome

Locate the tibiofibular joints and explain that the superior is a synovial joint whilst the inferior is fibrous, and that only slight movement occurs. ☐

Identify the (superficial and deep) muscles of the calf. ☐

Label a diagram of the fascial compartments of the lower limb, particularly in the context of blood vessels and nerves. ☐

Describe the fascial compartments enclosing the major muscle groups of the leg and explain the functional importance of these compartments and their contents in relation to compartment syndromes. ☐

Identify the bones of the ankle and foot on models and radiographs, and the ligaments stabilising the ankle joint. ☐

Recognise the muscles of the anterior compartment of the leg, their innervation and blood supply. ☐

Explain the factors involved in the stability of the ankle joint, particularly the lateral ligaments, and explain the injury in 'sprain'. ☐

Describe the arches of the foot and their importance, and explain how they are maintained by ligaments and muscles. ☐

Describe the movements of inversion and eversion at the subtalar joint, the muscles responsible, their innervations and main attachments. ☐

Trace the paths of the medial plantar artery and nerve and identify the structures they supply. ☐

Trace the paths of the lateral plantar artery and nerve and identify the structures they supply. ☐

Identify the plantar aponeurosis, which helps stabilise the foot medially. ☐

Identify the muscles of the sole of the foot. ☐

Session learning outcome:

To be able to identify the structures of the ankle and foot and link them to function.

Human Structure Learning Outcomes

Pelvis

Week 212

Reproductive

Learning outcome

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| Identify features of the bony pelvis and define the terms 'greater' and 'lesser' pelvis. | ☐ |
| Demonstrate the main palpable anatomical landmarks of the iliac, ischial and pubic bones in the living, on bones and suitable radiographs. | ☐ |
| Identify and describe the joints and ligaments of the pelvis. | ☐ |
| Identify the boundaries and major features of the pelvic inlet and the component parts of the pelvic outlet (recognising their normal orientation). | ☐ |
| Compare and contrast the female and male pelvis. | ☐ |
| Describe and identify the boundaries, divisions and structures within the perineum. | ☐ |
| Describe and identify the components of the anterior and posterior pelvic wall and pelvic floor. | ☐ |
| Demonstrate the points of attachment of the muscles of the abdominal wall and those of the levator ani. | ☐ |
| Discuss the functions of the pelvic floor musculature, its midline raphe and the structures passing through it in females. | ☐ |
| Describe and identify the location of the perineal membrane and the perineal body. | ☐ |
| Describe the origins, courses and relations of the uterine and ovarian arteries. | ☐ |
| Describe the paths and innervations of the pudendal nerves, and the sites that may be injected for anaesthesia during childbirth. | ☐ |
| Explain the function of the greater vestibular glands (Bartholin's glands) | ☐ |
| Identify the recto-uterine (pouch of Douglas) and the vesico-uterine pouch. | ☐ |
| Describe the lymphatic drainage of the pelvis. | ☐ |
| Interpret standard diagnostic images of the pelvis and be able to recognise common abnormalities. | ☐ |

Session learning outcome:

To be able to identify the musculoskeletal structures of the pelvis and perineum and link them to function.

Human Structure Learning Outcomes

Pulmonary Vessels

Week 213

Respiratory

Learning outcome

Describe the course of emboli (mobile blood clots) and the pathological effects in both the venous and arterial circulations. ☐

Define the term 'bronchopulmonary segment' and explain its importance. ☐

Identify parts of the pulmonary circulation to the second division of great vessels (very briefly). ☐

Describe the blood of the lungs. ☐

List the differences between the pulmonary circulation and the bronchial circulation, and recognise the links between them. ☐

Describe the basic conditions predisposing to thrombosis - Virchow's Triad. ☐

Be aware of the different types of embolus other than thrombus. ☐

Identify structures within the alveolar wall and relate them to function. ☐

Session learning outcome:

To be able to identify the circulatory vessels of the lung.

Human Structure Learning Outcomes

Face

Week 219

Nervous

Learning outcome

Identify in detail the nerves that carry cutaneous innervation of the skin of the face and scalp. ☐

Explain that the face has no deep fascia, only superficial fascia. ☐

Recognise the extracranial courses of the branches of the facial nerve. ☐

Link the muscles of facial expression to each branch of the facial nerve and describe the consequences of injury to each branch. ☐

Name the facial muscles of expression as either sphincters or dilators by describing their actions (and innervations). ☐

Describe the intracranial and intrapetrous course of the facial nerve and the relationships of its major branches to the middle ear in relation to damage of the nerve within the facial canal. ☐

Describe the origin, functions and major branches of the sensory and motor components of the facial nerve. ☐

Describe the facial artery and transverse facial artery, their origins and courses. ☐

Describe the facial veins and their tributaries, explaining the risks of the spread of infection intracranially. ☐

Describe the relationship of the termination of the facial vein (draining into the internal jugular vein) and the mandibular branch of the retromandibular vein (draining facial muscles controlling the angle of the mouth) to the submandibular gland and related upper jugular lymph nodes in relation to exploration of this area. ☐

Describe the anatomy of the scalp, naming its individual layers. ☐

Describe the blood supply of the scalp and its significance in laceration injuries. ☐

Locate the occipitofrontalis muscle and describe its actions. ☐

Trace the path and distribution of the maxillary artery and its main branches in the infratemporal fossa. ☐

Describe the pterygoid venous plexus. ☐

Recognise structures of the face on radiographic images. ☐

Session learning outcome:

To be able to identify the structures of the face and scalp.

Human Structure Learning Outcomes

Elbow And Upper Limb Vasculature

Week 220

Renal

Learning outcome

Recognise the parts of the elbow joint with respect to bones and connective tissues. ☐

Demonstrate the movements of flexion and extension at the elbow joint, identifying the muscles involved whilst summarising their main attachments and somatic nervous innervation. ☐

Define the boundaries of the cubital fossa and the relationships of the structures therein. ☐

Locate the muscles biceps brachii, brachialis, coracobrachialis, triceps brachii and anconeus, and describe them as muscle groups as defined by fascial compartments, and recognise the functional significance of these groups. ☐

Trace the origins, courses and distributions of the major arteries and their branches that supply the shoulder, arm and forearm in relation to common sites of injury. ☐

Explain the importance of anastomoses between branches of the arteries at the shoulder and in the upper limb. ☐

Demonstrate the locations at which the pulses in the brachial, radial and ulnar arteries may be located. ☐

Describe the courses of the main veins of the upper limb; classify and contrast the functions of the deep and superficial veins. ☐

Recognise the common sites of venous access (with key relations). ☐

Explain the relationship between venous and lymphatic drainage channels. ☐

Identify and side a radius, and identify its main anatomical landmarks. ☐

Identify and side an ulna, and identify its main anatomical landmarks. ☐

Describe the motor and sensory innervations (of the hand), and tests appropriate to the radial, median and ulnar nerves. ☐

Identify the collateral ligament of the elbow joint, the annular ligament, the triangular fibrocartilage of inferior radio-ulnar joint, the interosseus membrane in the forearm and the collateral ligaments of the joints of digits. ☐

Describe the close relations of the bones and joints of the upper limb (e.g. bursae, blood vessels, nerves, ligaments and tendons), which may be injured in fractures or dislocations and predict what the functional effects of such damage would be. ☐

Describe the anatomy of the superior radio-ulnar joints. ☐

Session learning outcome:

To be able to identify the musculoskeletal and neurovascular structures of the upper limb and link them to function.

Human Structure Learning Outcomes

Meninges & Ventricles

Week 221

Nervous

Learning outcome

Identify, describe and label the three layers of meninges: the dura mater, the arachnoid mater, and the pia mater and the important structures relating to them. ☐

Describe the endosteal and meningeal layers of the dura mater, the four septa created by it dividing up the cranial cavity, and the relationship with the major venous sinuses (falx cerebri, tentorium cerebelli, falx cerebelli, diaphragma sellae). ☐

Describe the anatomical structures encountered during the following procedures: lumbar puncture, spinal (subarachnoid) anaesthetic block, epidural (peridural) anaesthetic block. Explain the indications and risks associated with these techniques. ☐

Identify the components of the ventricular system and link to the production and circulation of cerebrospinal fluid and the functional consequences of obstruction of cerebrospinal fluid drainage. ☐

Session learning outcome:

To be able to identify the cranial nerves, meningeal layers and the main components of the ventricular system.

Human Structure Learning Outcomes

Skin

Week 227

Musculoskeletal

Learning outcome

To be able to identify the layers of the skin histologically and link with function.



Session learning outcome:

To be able to identify the layers of the skin histologically and link with function.

Human Structure Learning Outcomes

Breast Anatomy

Week 228

Reproductive

Learning outcome

Describe the gross structure of the breast, its blood supply and changes with puberty/pregnancy/menopause. ☐

Describe the anatomy of the axillary lymph nodes and explain their importance in the lymphatic drainage of the breast and the skin of the trunk and upper limb and in the spread of tumours. ☐

Recognise structures of the breast on mammography, ultrasound and MRI imaging. ☐

Session learning outcome:

To be able to identify the anatomy of the breast and recall the lymphatic drainage of the breast, skin of the trunk and upper limb.

Human Structure Learning Outcomes

Liver And Spleen

Week 229

Gastrointestinal

Learning outcome

Identify the lobes and ligaments of the liver, their anatomical relations, the peritoneal reflections and movements during respiration. ☐

Identify at the porta hepatis (the hilum of the liver): the hepatic artery, the hepatic portal vein, and the hepatic ducts. ☐

Summarise the functional anatomy of the portal vein, the portal venous system and portal-systemic anastomoses and their significance in portal hypertension. ☐

Identify the position (in relation to the ribs) and form of the spleen in relation to its palpation through the abdominal wall and its key anatomical relationships with other abdominal structures. ☐

Explain the significance of the anatomical relationships of the spleen in relation to trauma, chronic infections and disorders of the haematopoietic system. ☐

Identify the splenic artery and vein. ☐

Identify the peritoneum surrounding the spleen and the two ligaments it forms (lienorenal and gastrosplenic). ☐

Describe the anatomy of the attachments, openings and relationships of the diaphragm, the surface anatomy and the movements it makes during breathing. ☐

Identify the structures that pass through the diaphragm. ☐

Explain referred pain from the diaphragm and its underlying peritoneum. ☐

Identify and describe the locations of the 8 segments of the liver. ☐

Session learning outcome:

To be able to identify the liver and spleen, their anatomical relationships and link them to function and common pathology.

Human Structure Learning Outcomes

Neuroanatomy

Week 230

Nervous

Learning outcome

Identify the basal ganglia and their main functions, and connections to other parts of the central nervous system. ☐

Identify the thalamus and its main functions and connections to other parts of the central nervous system. ☐

Identify the components of the limbic system, their main functions, and connections to other parts of the central nervous system. ☐

Session learning outcome:

To be able to identify and list the main functions of the basal ganglia, thalamus and limbic system and identify the cellular components and histopathological hallmarks of the brain and peripheral nervous system.

Human Structure Learning Outcomes

Spinal Tracts

Week 231

Nervous

Learning outcome

Identify the major tracts of the brainstem in transverse section. ☐

Identify the pathway of the corticospinal tract from the pre-central gyrus to the anterior horn cell and link to the functional consequences of damage at different points along that pathway. ☐

Identify the routes of the two principal sensory pathways from the dorsal root ganglion to the thalamus and parietal lobe and link to the functional consequences of damage at different points along that pathway. ☐

Recognise the normal brainstem and cerebellum, and common pathology on radiological images. ☐

Session learning outcome:

To be able to identify the main components of the cerebellum and its connections to the brain stem, spinal cord and cerebral hemispheres.

Human Structure Learning Outcomes

Cardiovascular & Respiratory Anatomy & Pathology

Week 237

Musculoskeletal

Learning outcome

Predict the effects of pathology on the anatomy of the cardiovascular and respiratory system.



Session learning outcome:

To predict the effects of pathology on the anatomy of the cardiovascular and respiratory system.

Human Structure Learning Outcomes

Gastrointestinal Tract Anatomy & Pathology

Week 238

Endocrine

Learning outcome

Predict the effects of pathology on the anatomy of the gastrointestinal tract.



Session learning outcome:

To predict the effects of pathology on the anatomy of the GI tract.

Human Structure Learning Outcomes

Renal & Reproductive Anatomy & Pathology

Week 239

Nervous

Learning outcome

Predict the effects of pathology on the anatomy of the renal system. ☐

Predict the effects of pathology on the anatomy of the reproductive system. ☐

Session learning outcome:

To predict the effects of pathology on the anatomy of the urinary system. To predict the effects of pathology on the anatomy of the reproductive system.

Human Structure Learning Outcomes

Neuroanatomy & Pathology

Week 240

Reproductive

Learning outcome

Identify histopathological hallmarks of Alzheimer's disease, Lewy bodies and vascular dementia.



Session learning outcome:

To predict the effects of pathology on the anatomy of the nervous system and identify histopathological hallmarks of Alzheimers disease.